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DEVELOPING APPARATUS AND IMAGE FORMING APPARATUS

Abstract

A developing apparatus comprises a development chamber with a developer bearing member and supply member, and a developer accommodation chamber above it. A stirring member with a sheet member rotates to stir developer. All members rotate along the longitudinal direction, with an opening along the chamber walls. In cross-section, the opening has upstream and downstream ends relative to the stirring member's rotation. At the longitudinal center, the distance from the upstream end to the contact region between the developer bearing member and supply member is shorter than the same distance measured at the longitudinal end.

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

[0001] The present invention relates to a developing apparatus and an image forming apparatus.

Description of the Related Art

[0002] An image forming apparatus for forming an image on a recording medium using an electrophotographic image forming system is known. Examples of the image forming apparatus may include an electrophotographic copier, an electrophotographic printer (such as a LED printer or a laser beam printer), a facsimile device, and a word processor.

[0003] Japanese Patent Application Publication No. 2001-194883 discloses a development unit configuration having a development chamber in which a development roller and a supply roller are set, and a developer accommodation chamber for accommodating a developer (toner) therein. In Japanese Patent Application Publication No. 2001-194883, in the developer accommodation chamber provided at the top of the development chamber, a stirring member for stirring a developer is set.

[0004] When in the development unit as in Japanese Patent Application Publication No. 2001-194883, or a process cartridge including the same, the developer amount is increased due to an increase in volume, or a developer having a heavy specific gravity is used, the developer in the developer accommodation chamber becomes more likely to be pushed into the development chamber set at the lower part due to its own weight of the developer and the movement of the stirring member, resulting in a dense state of the developer in the development chamber. As a result, the poor circulation of the developer in the development chamber is caused, so that the functions of the development roller and the supply roller are impaired. This may result in the occurrence of a harmful effect on an image such as toner leakage. Under such circumstances, conventionally, the developer accommodation chamber has been set in a lying posture in the direction horizontal with the development chamber, thereby preventing its own weight of the developer from being imposed. However, when the developer accommodation chamber is set in a lying posture, an increase in size of the apparatus may be caused.

SUMMARY OF THE INVENTION

[0005] The present invention was completed in view of the foregoing problem. It is an object of the present invention to circulate a developer in a development unit properly.

[0006] The present invention provides a developing apparatus configured to comprise: [0007] a developer bearing member for bearing a developer; [0008] a supply member for coming in contact with the developer bearing member, thereby forming a contact region, and supplying the developer to a surface of the developer bearing member in the contact region; [0009] a development chamber including the developer bearing member and the supply member arranged therein; [0010] a developer accommodation chamber for accommodating the developer; and [0011] a stirring member having a shaft, and a sheet member held by the shaft, the sheet member stirring the developer accommodated in the developer accommodation chamber by rotation of the shaft, wherein, [0012] the developer accommodation chamber is set at a top in a gravity direction of the development chamber in a posture in which the developing apparatus is mounted in an image forming apparatus, [0013] the developer bearing member, the supply member, and the stirring member are rotatable around a rotation axis along a longitudinal direction of the developing apparatus as a center, [0014] a wall surface portion for partitioning the developer accommodation chamber and the development chamber is provided with an opening for enabling a movement of the developer from the developer accommodation chamber to the development chamber along the longitudinal direction, and [0015] in a cross section in the longitudinal direction, the opening has

an end on an upstream side and an end on a downstream side in a direction of passage of the sheet member according to rotation of the stirring member, and a distance between a first end and a contact region of the developer bearing member and the supply member is shorter than a distance between a second end and the contact region where the first end represents the end on the upstream side at a central portion in the longitudinal direction, and the second end represents the end on the upstream side at the end in the longitudinal direction.

[0016] The present invention also provides an image forming apparatus configured to comprise:

[0017] a developer bearing member for bearing a developer; [0018] a supply member for coming in contact with the developer bearing member, thereby forming a contact region, and supplying the developer to a surface of the developer bearing member in the contact region; [0019] a development chamber including the developer bearing member and the supply member arranged therein; [0020] a developer accommodation chamber for accommodating the developer; [0021] a stirring member having a shaft, and a sheet member held by the shaft, the sheet member stirring the developer accommodated in the developer accommodation chamber by rotation of the shaft; and [0022] an image bearing member for developing an electrostatic latent image formed on a surface when the developer is supplied from the developer bearing member,

wherein, [0023] the developer accommodation chamber is set at a top in a gravity direction of the development chamber, [0024] the developer bearing member, the supply member, and the stirring member are each rotatable around a rotation axis along a longitudinal direction as a center, [0025] a wall surface portion for partitioning the developer accommodation chamber and the development chamber is provided with an opening for enabling a movement of the developer from the developer accommodation chamber to the development chamber along the longitudinal direction, and [0026] in a cross section in the longitudinal direction, the opening has an end on an upstream side and an end on a downstream side in a direction of passage of the sheet member according to rotation of the stirring member, and a distance between a first end and a contact region of the developer bearing member and the supply member is shorter than a distance between a second end and the contact region where the first end represents the end on the upstream side at a central portion in the longitudinal direction, and the second end represents the end on the upstream side at the end in the longitudinal direction.

[0027] The present invention also provides a developing apparatus configured to comprise: [0028] a developer bearing member for bearing a developer; [0029] a supply member for coming in contact with the developer bearing member, thereby forming a contact region, and supplying the developer to a surface of the developer bearing member in the contact region; [0030] a development chamber including the developer bearing member and the supply member arranged therein; [0031] a developer accommodation chamber for accommodating the developer; and [0032] a stirring member having a shaft, and a sheet member held by the shaft, the sheet member stirring the developer accommodated in the developer accommodation chamber by rotation of the shaft, wherein, [0033] the developer accommodation chamber is set at a top in a gravity direction of the development chamber in a posture in which the developing apparatus is mounted in an image forming apparatus, [0034] the developer bearing member, the supply member, and the stirring member are each rotatable around a rotation axis along a longitudinal direction of the developing apparatus as a center, [0035] a wall surface portion for partitioning the developer accommodation chamber and the development chamber is provided with an opening for enabling a movement of the developer from the developer accommodation chamber to the development chamber along the longitudinal direction, and [0036] in a cross section in the longitudinal direction, the opening has an end on an upstream side and an end on a downstream side in a direction of passage of the sheet member according to rotation of the stirring member, and a width between the end on the upstream side and the end on the downstream side at a central portion in the longitudinal direction is narrower than a width between the end on the upstream side and the end on the downstream side at the ends in the longitudinal direction.

[0037] In accordance with a configuration of the present invention, it is possible to circulate the developer in a development unit properly.

[0038] Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0039] FIG. 1A is a cross sectional view showing a shape of an opening of a development unit;

[0040] FIG. 1B is a cross sectional view showing a shape of an opening of a development unit;

[0041] FIG. 2 is a schematic cross sectional view showing a configuration of a printer;

[0042] FIG. 3 is a perspective view of a process cartridge and a toner cartridge;

[0043] FIG. 4 is a perspective view of a process cartridge and a toner cartridge;

[0044] FIG. 5 is a side view of a process cartridge and a toner cartridge;

[0045] FIG. 6 is a side view of a process cartridge and a toner cartridge;

[0046] FIG. 7 is a cross sectional view of a process cartridge and a toner cartridge;

[0047] FIG. 8 is a cross sectional view of a process cartridge and a toner cartridge;

[0048] FIG. 9 is a side view for illustrating a driving transmission path of a process cartridge;

[0049] FIG. 10 is a view of a process cartridge and a toner cartridge as seen in the removing direction;

[0050] FIG. 11A is a perspective view of a toner cartridge;

[0051] FIG. 11B is a perspective view of a toner cartridge;

[0052] FIG. 12A is a perspective view of a toner cartridge;

[0053] FIG. 12B is a perspective view of a toner cartridge;

[0054] FIG. 13 is a view of a toner cartridge as seen in the removing direction;

[0055] FIG. 14 is a view showing an inside structure of a toner cartridge;

[0056] FIG. 15 is a cross sectional view of a toner cartridge;

[0057] FIG. 16A is a side view of a toner cartridge;

[0058] FIG. 16B is a side view of a toner cartridge;

[0059] FIG. 17 is a side view of a toner cartridge;

[0060] FIG. 18A is a cross sectional view showing a position and a shape of an opening of a development unit;

[0061] FIG. 18B is a cross sectional view showing a position and a shape of an opening of a development unit;

[0062] FIG. 19A is a cross sectional view showing the movement of a developer in a process cartridge;

[0063] FIG. 19B is a cross sectional view showing a movement of a developer in a process cartridge;

[0064] FIG. 20 is a cross sectional view showing a shape of an opening of a development unit;

[0065] FIG. 21A is a cross sectional view showing a shape of an opening of a development unit in accordance with another Example;

[0066] FIG. 21B is a cross sectional view showing a shape of an opening of a development unit in accordance with a still other Example;

[0067] FIG. 22A is a cross sectional view showing a shape of an opening of a development unit;

[0068] FIG. 22B is a cross sectional view showing a shape of an opening of a development unit; and

[0069] FIG. 23 is a block view showing a control configuration of an image forming apparatus.

DESCRIPTION OF THE EMBODIMENTS

[0070] Below, referring to the accompanying drawings, preferred embodiments of the present

invention will be described exemplarily in details. However, the dimensions, the materials, and the shapes of the constituent components described in the following embodiments, the relative arrangement thereof, and the like should be appropriately changed according to the configuration of the apparatus to which the present invention is applied, and various conditions. Therefore, it is not intended that the scope of the present invention is limited only thereto unless otherwise specified.

EXAMPLE 1

Whole Outline of Printer

[0071] The basic configuration and the operation of a printer **100** as an image forming apparatus in accordance with the present Example will be described by reference to FIG. 2. FIG. 2 is a schematic cross sectional view showing the configuration of the printer **100** in accordance with the present Example. An arrow Z indicates the vertical direction, and an arrow H indicates a horizontal direction. As shown in FIG. 2, the printer **100** has an apparatus main body **100A**, and a process cartridge P and a toner cartridge T removably mounted in the apparatus main body **100A** as cartridges.

[0072] The apparatus main body **100A** has a scanner **101** as an exposure device, a loading tray **102** for loading sheets S thereon, a paper feed roller **103**, a transfer roller **104**, a fixing portion **105**, a discharge tray **106**, and a control portion **107**. Further, the process cartridge P and the toner cartridge T (developer cartridges) are removably mounted in the apparatus main body **100A**.

[0073] The process cartridge P has a photosensitive drum **12**, a charging roller **13** (charging member), a cleaning blade **14** (cleaning member), a drum unit **10**, and a development unit **30** (developing apparatus) having a development roller **32**, a supply roller **33**, and a development frame body **31**. A further configuration of the process cartridge P will be described later. The drum unit **10** has a drum frame body **11** described later. The photosensitive drum **12** is rotatably supported by the drum frame body **11**. The development roller **32** is rotatably supported by the development frame body **31**. Incidentally, a configuration in which the drum cartridge corresponding to the drum unit **10**, and the development cartridge corresponding to the development unit **30** are respectively detachably set with respect to the image forming apparatus main body is also acceptable in place of such a configuration that the process cartridge P includes the drum unit **10** and the development unit **30**.

[0074] The toner cartridge T can be mounted with respect to the process cartridge P. The toner cartridge T is configured so as to accommodate a toner as a developer, and to supply the toner to the development unit **30** of the process cartridge P. Although will be described by reference to the accompanying drawings later, the toner cartridge T has a toner transport member **62**, a toner transport screw **63**, and a toner frame body **55**. The toner transport member **62** and the toner transport screw **63** are rotatably supported by the toner frame body **55**.

[0075] The image formation operation onto a sheet S will be described. The control portion **107** of the printer **100** starts the image formation operation onto the sheet S on the basis of a signal received from an external device **200** shown in FIG. 23. The control portion **107** includes an information processing device having arithmetic resources such as a processor and a memory, a processing circuit, and the like. As shown in a block view of FIG. 23, the control portion **107** acquires information from each constituent element of the printer **100** according to the program stored in a memory, or an user instruction using the external device **200**, and performs control by a control signal. The printer **100** includes a driving source **160** having a motor, or the like, and a power supply portion **170** for supplying an electric power such as a driving voltage to each constituent element.

[0076] First, the photosensitive drum **12** is rotated by the driving source **160** of the apparatus main body **100A**. The charging roller **13** is rotated following the photosensitive drum **12** with the charging roller **13** applied with a charging voltage by the power supply portion **170**. As a result, the surface of the photosensitive drum **12** is uniformly charged. On the basis of image information, the scanner **101** applies a laser toward the charged surface of the photosensitive drum **12**, so that an

electrostatic latent image is formed on the surface of the photosensitive drum **12**.

[0077] When the development roller **32** is applied with a development voltage by the power supply portion **170**, a toner is supplied to the photosensitive drum **12** from the development roller **32**. As a result, an electrostatic latent image is developed as a toner image. Subsequently, the rotation of the photosensitive drum **12** transports the toner image formed at the photosensitive drum **12** toward the transfer portion formed between the transfer roller **104** and the photosensitive drum **12**.

[0078] On the other hand, a sheet S is fed from the loading tray **102** by the paper feed roller **103**. The sheet S is transported to the transfer portion in accordance with the timing at which the toner image formed at the photosensitive drum **12** reaches the transfer portion.

[0079] A transfer voltage is applied to the transfer roller **104** from the power supply portion **170**, so that the toner image is transferred from the photosensitive drum **12** to the sheet S. The residual toner not transferred to the sheet S is removed from the surface of the photosensitive drum **12** by a cleaning blade **14**.

[0080] The sheet S onto which the toner image has been transferred is transported toward the fixing portion **105**. When the sheet S passes through the fixing portion **105**, the toner image is heated and pressurized by a fixing portion **105**, thereby to be fixed onto the sheet S.

[0081] The printer **100** in accordance with the present Example can carry out double-sided printing in which an image formation operation is performed on the front surface and the back surface of the sheet S. When an image is formed on only the front surface of the sheet S, the sheet S that has passed through the fixing portion **105** is discharged to the discharge tray **106**. On the other hand, when double-side printing is carried out, the sheet S with the toner image fixed on the surface thereof is transported in an arrow M direction through a double-side transport path **81**, and reaches the transfer portion again. As a result, a toner image is formed on the back surface of the sheet S. Subsequently, the sheet S passes through the fixing portion **105**, and is discharged to the discharge tray **106**.

[0082] Attachment and Removal of Process Cartridge and Toner Cartridge

[0083] Referring to FIGS. **2**, **3**, **4**, **5**, **6**, and **10**, a description will be given to attachment and removal of a process cartridge P and a toner cartridge T in accordance with the present Example with respect to the apparatus main body **100A**.

[0084] FIGS. **3** and **4** are each a perspective view of the process cartridge P and the toner cartridge T. FIGS. **5** and **6** are each a side view of the process cartridge P and the toner cartridge T. The photosensitive drum **12** is rotatable around the rotation axis (first axis) **12a** shown in FIG. **6**. The direction of extension of the rotation axis **12a** is referred to as the rotation axis direction (axis direction) of the photosensitive drum **12**.

[0085] FIG. **3** is a perspective view of the process cartridge P and the toner cartridge T as seen from the driving side. FIG. **4** is a perspective view of the process cartridge P and the toner cartridge T as seen from a non-driving side. FIG. **5** is a side view of the driving side of the process cartridge P and the toner cartridge T as seen in the rotation axis direction of the photosensitive drum **12**. FIG. **6** is a side view of the non-driving side of the process cartridge P and the toner cartridge T as seen in the rotation axis direction of the photosensitive drum **12**.

[0086] As shown in FIG. **2**, the printer **100** has a door (opening/closing member) **100B** that covers an opening **100C** of the apparatus main body **100A**. The door **100B** is attached rotatably around the rotation axis **100D** with respect to the apparatus main body **100A**. The door **100B** is configured movably in an arrow J direction from the closing position covering the opening **100C** to the opening position exposing the opening **100C**. With the door **100B** present at the opening position, the attachment of the process cartridge P and the toner cartridge T to the apparatus main body **100A**, and the removal of the process cartridge P and the toner cartridge T from the apparatus main body **100A** through the opening **100C** are allowed.

[0087] FIG. **10** is a view of the process cartridge P and the toner cartridge T as seen in the removal direction PDD. As shown in FIG. **10**, with regard to the direction of the rotation axis **12a** of the

photosensitive drum **12**, the drum frame body **11** has a process driving end (the first end of the drum frame body **11**) **11f1**, and a process non-driving end (the second end of the drum frame body **11**) **11f2** opposite to the process driving end **11f1**. The process driving end **11f1** and the process non-driving end **11f2** are the portions (ends) situated at the outermost positions in the drum frame body **11** in the rotation axis direction of the photosensitive drum **12**. A plurality of the process driving ends **11f1** and a plurality of the process non-driving ends **11f2** may be present. The center of the drum frame body **11** in the rotation axis direction of the photosensitive drum **12** is referred to as a center **11f3**. The distance between the center **11f3** and the process driving end **11f1** of the drum frame body **11** and the distance between the center **11f3** and the process non-driving end **11f2** thereof are equal to each other.

[0088] In the present Example, the process driving end **11f1** and the process non-driving end **11f2** are portions (ends) situated at the outermost positions of the process cartridge P in the rotation axis direction of the photosensitive drum **12**. In other words, in the rotation axis direction of the photosensitive drum **12**, the process driving end **11f1** and the process non-driving end **11f2** are in agreement with the driving end (the first end of the process cartridge P) and the non-driving end (the second end of the process cartridge P) of the process cartridge P. In the rotation axis direction of the photosensitive drum **12**, the “driving side” of the drum frame body **11** or the process cartridge P is situated opposite to the “non-driving side” of the drum frame body **11** or the process cartridge P.

[0089] In the rotation axis direction of the photosensitive drum **12**, the side on which the process driving end **11f1** is arranged with respect to the center **11f3** of the drum frame body **11** is the driving side of the drum frame body **11** or the driving side of the process cartridge P. In the rotation axis direction of the photosensitive drum **12**, the side on which the process non-driving end **11f2** is arranged with respect to the center **11f3** of the drum frame body **11** is the non-driving side of the drum frame body **11** or the non-driving side of the process cartridge P. In the present Example, in the rotation axis direction of the photosensitive drum **12**, the center **11f3** of the drum frame body **11** is the same as the center of the process cartridge P.

[0090] Although described later, the toner transport member **62** shown in FIG. **14** is rotatable around the rotation axis **62a**. The direction of extension of the rotation axis **62a** is referred to as the rotation axis direction (axis direction) of the toner transport member **62**. Further, the toner transport screw **63** shown in FIG. **14** is rotatable around the rotation axis **63a**. The direction of extension of the rotation axis **63a** is referred to as the rotation axis direction (axis direction) of the toner transport screw **63**.

[0091] As shown in FIG. **10**, with regard to the rotation axis direction LD of the toner transport screw **63**, the toner frame body **55** has a toner driving end (the first end of the toner frame body **55**) **55a1**, and a toner non-driving end **55a2** (the second end of the toner frame body **55**) opposite to the toner driving end **55a1**. The toner driving end **55a1** and the toner non-driving end **55a2** are portions (ends) situated at the outermost positions in the toner frame body **55** in the rotation axis direction of the toner transport screw **63**. A plurality of the toner driving ends **55a1** and a plurality of the toner non-driving ends **55a2** may be present. In the rotation axis direction of the toner transport screw **63**, the center of the toner frame body **55** is referred to as a center **55a3**. The distance between the center **55a3** and the toner driving end **55a1** of the toner frame body **55** and the distance between the center **11f3** and the toner non-driving end **55a2** thereof are equal to each other.

[0092] In the present Example, the toner driving end **55a1** and the toner non-driving end **55a2** are portions (ends) situated at the outermost position in the toner cartridge T in the rotation axis direction of the toner transport screw **63**. In other words, in the rotation axis direction of the toner transport screw **63**, the toner driving end **55a1** and the toner non-driving end **55a2** are in agreement with the driving end (the first end of the toner cartridge T) and the non-driving end (the second end of the toner cartridge T) of the toner cartridge T.

[0093] In the rotation axis direction of the toner transport screw **63**, the side on which the toner driving end **55a1** is arranged with respect to the center **55a3** of the toner frame body **55** is the driving side of the toner frame body **55**, or the driving side of the toner cartridge T. In the rotation axis direction of the toner transport screw **63**, the side on which the toner non-driving end **55a2** is arranged with respect to the center **55a3** of the toner frame body **55** is the non-driving side of the toner frame body **55**, or the non-driving side of the toner cartridge T. In the present Example, in the rotation axis direction of the toner transport screw **63**, the center **55a3** of the toner frame body **55** is the same as that of the toner cartridge T.

[0094] In the rotation axis direction of the toner transport screw **63**, the driving side of the toner frame body **55** or the driving side of the toner cartridge T is situated opposite to the non-driving side of the toner frame body **55** or the non-driving side of the toner cartridge T.

[0095] In the present Example, the rotation axis direction of the photosensitive drum **12**, the rotation axis direction of the toner transport member **62**, and the rotation axis direction of the toner transport screw **63** are in parallel to one another. Therefore, the rotation axis direction of the photosensitive drum **12**, the rotation axis direction of the toner transport member **62**, and the rotation axis direction of the toner transport screw **63** are each also referred to simply as the axis direction (the first direction) LD.

[0096] In the present Example, in the axis direction LD, the position of the center **55a3** of the toner frame body **55** and the position of the center **11f3** of the drum frame body **11** are the same.

However, the position of the center **55a3** of the toner frame body **55** and the position of the center **11f3** of the drum frame body **11** may be different from each other.

[0097] As shown in FIGS. **3** and **5**, the process cartridge P has a driving side process guide **22** on the driving side of the drum frame body **11**. The toner cartridge T has a driving side toner guide **51** on the driving side of the toner frame body **55**. As shown in FIGS. **4** and **6**, the process cartridge P has a non-driving side process guide **23** on the non-driving side of the drum frame body **11**. The toner cartridge T has a non-driving side toner guide **52** on the non-driving side of the toner frame body **55**.

[0098] The direction in which the process cartridge P is attached with respect to the apparatus main body **100A** is referred to as the attachment direction PDA. The direction in which the process cartridge P is removed from the apparatus main body **100A** is referred to as the removal direction PDD. The attachment direction PDA and the removal direction PDD are collectively referred to as the removal/attachment direction PD of the process cartridge P. The driving side process guide **22** and the non-driving side process guide **23** are formed along the removal/attachment direction PD. The driving side process guide **22** and the non-driving side process guide **23** are guided by a guide portion of the apparatus main body **100A**, and the process cartridge P moves in the removal/attachment direction PD with respect to the apparatus main body **100A**.

[0099] The direction in which the toner cartridge T is attached to the apparatus main body **100A** is referred to as an attachment direction TDA. The direction in which the toner cartridge T is removed from the apparatus main body **100A** is referred to as a removal direction TDD. The attachment direction TDA and the removal direction TDD are collectively referred to as a removal/attachment direction TD of the toner cartridge T. The driving side toner guide **51** and the non-driving side toner guide **52** are formed along the removal/attachment direction TD. The driving side toner guide **51** and the non-driving side toner guide **52** are guided by the guide portion of the apparatus main body **100A**, and the toner cartridge T moves in the removal/attachment direction TD with respect to the apparatus main body **100A**.

[0100] In the present Example, the removal/attachment direction PD of the process cartridge P is the direction crossing the axis direction LD. The angle formed between the direction orthogonal to the axis direction LD and the removal/attachment direction PD is preferably smaller than the angle formed between the axis direction LD and the removal/attachment direction PD, and the removal/attachment direction PD is more preferably the direction orthogonal to the axis direction

LD.

[0101] In the present Example, the removal/attachment direction TD of the toner cartridge T is the direction crossing the axis direction LD. The angle formed between the direction orthogonal to the axis direction LD and the removal/attachment direction TD is preferably smaller than the angle formed between the axis direction LD and the removal/attachment direction TD, and the removal/attachment direction TD is more preferably the direction orthogonal to the axis direction LD.

[0102] Incidentally, in the present Example, although the removal/attachment direction PD and the removal/attachment direction TD are in parallel with each other, the removal/attachment direction PD and the removal/attachment direction TD may be different from each other.

[0103] In the present Example, the attachment and the removal of the process cartridge P are performed with the toner cartridge T not attached to the apparatus main body **100A**. In other words, the attachment and the removal of the process cartridge P are performed before the attachment of the toner cartridge T with respect to the apparatus main body **100A**.

[0104] With the toner cartridge T not attached with respect to the apparatus main body **100A**, the process cartridge P is attached with respect to the apparatus main body **100A** through the opening **100C**. Further, with the process cartridge P attached with respect to the apparatus main body **100A**, the toner cartridge T is attached with respect to the apparatus main body **100A** and the process cartridge P through the opening **100C**.

[0105] With the toner cartridge T and the process cartridge P attached with respect to the apparatus main body **100A**, in the attachment direction PDA and the attachment direction TDA, the process cartridge P is situated on the downstream side of the toner cartridge T, and the toner cartridge T is situated on the upstream side of the process cartridge P.

[0106] When the toner cartridge T and the process cartridge P are removed from the apparatus main body **100A**, the toner cartridge T is removed from the apparatus main body **100A** and the process cartridge P through the opening **100C**. Subsequently, the process cartridge P is removed from the apparatus main body **100A** through the opening **100C**.

Toner Cartridge

[0107] Referring to FIGS. **7**, **8**, and **11A** and **11B** to **17**, a configuration of the toner cartridge T in accordance with the present Example will be described in further details.

[0108] FIGS. **11A**, **11B**, **12A**, and **12B** are each a perspective view of the toner cartridge T. Specifically, FIGS. **11A** and **11B** are each a perspective view of the toner cartridge T as seen from the driving side. In FIG. **11B**, a part of the toner cartridge T is omitted. FIGS. **12A** and **12B** are each a perspective view of the toner cartridge T as seen from the non-driving side. In FIG. **12B**, a part of the toner cartridge T is omitted.

[0109] FIG. **13** is a view of the toner cartridge T as seen in the removal direction TDD. FIG. **14** is a view showing an inside structure of the toner cartridge T. FIG. **15** is a cross sectional view of the toner cartridge T with the direction orthogonal to the axis direction LD as the cross sectional direction.

[0110] FIGS. **16A**, **16B**, and **17** are each a side view of the toner cartridge T. Specifically, FIGS. **16A** and **16B** are each a side view of the driving side of the toner cartridge T as seen in the axis direction LD. FIG. **17** is a side view of the non-driving side of the toner cartridge T as seen in the axis direction LD.

[0111] As shown in FIGS. **7** and **14**, the toner frame body **55** of the toner cartridge T is provided with a toner accommodation chamber **53** (a first chamber) and a toner collecting chamber **54** (a second chamber). The toner to be supplied to the process cartridge P is accommodated in the toner accommodation chamber **53**. The toner returned from the process cartridge P by a return screw **18** shown in FIG. **8** is accommodated in the toner collecting chamber **54**.

[0112] The toner collecting chamber **54** and the toner accommodation chamber **53** are separated (isolated) from each other. Specifically, the toner frame body **55** has a partition wall **55b**, and the

toner collecting chamber **54** and the toner accommodation chamber **53** are completely partitioned by the partition wall **55b**. This prevents the toner from moving between the toner collecting chamber **54** and the toner accommodation chamber **53**, so that the toner accommodated in the toner collecting chamber **54** and the toner accommodated in the toner accommodation chamber **53** are prevented from being mixed with each other.

[0113] The volume of the toner accommodation chamber **53** is larger than the volume of the toner collecting chamber **54**. In the present Example, in the axis direction LD, the distance between the partition wall **55b** and the toner non-driving end **55a2** is shorter than the distance between the partition wall **55b** and the toner driving end **55a1**.

[0114] In the axis direction, the toner accommodation chamber **53** is arranged so as to overlap the center **55a3** of the toner frame body **55**. On the other hand, in the axis direction LD, the toner collecting chamber **54** is arranged on the non-driving side of the toner frame body **55**. In the axis direction LD, the distance between the toner collecting chamber **54** and the toner non-driving end **55a2** is shorter than the distance between the toner collecting chamber **54** and the toner driving end **55a1**. Further, in the axis direction LD, the toner collecting chamber **54** is arranged between the toner accommodation chamber **53** and the toner non-driving end **55a2**.

[0115] As shown in FIGS. **13** and **14**, the toner frame body **55** has a container portion **56** including the toner accommodation chamber **53** and the toner collecting chamber **54**, a driving end cover **57** arranged on the driving side of the toner cartridge T, and a non-driving end cover **58** arranged on the non-driving side of the toner cartridge T.

[0116] The container portion **56** includes a first container **56a** provided with the toner accommodation chamber **53**, and a second container **56b** provided with the toner collecting chamber **54**. In the present Example in which the second container **56b** is fixed to the first container **56a**, the sidewall of the first container **56a** and the sidewall of the second container **56b** are opposed to each other, and each have a function as the partition wall **55b**.

[0117] As shown in FIGS. **11A** and **11B**, the toner frame body **55** has a toner discharge port (a first opening) **61** for discharging the toner accommodated in the toner accommodation chamber **53** toward the development unit **30** of the process cartridge P. The toner discharge port **61** establishes a communication between the outside of the toner frame body **55** and the toner accommodation chamber **53**, so that the toner accommodated in the toner accommodation chamber **53** is discharged to the outside of the toner frame body **55** through the toner discharge port **61**. The toner discharge port **61** is included in the first container **56a**.

[0118] In the axis direction LD, the toner discharge port **61** is arranged closer to the driving side of the toner frame body **55**. In the axis direction LD, the distance between the toner discharge port **61** and the toner driving end **55a1** is shorter than the distance between the toner discharge port **61** and the toner non-driving end **55a2**.

[0119] The toner discharge port **61** is covered with a discharge port shutter not shown. The discharge port shutter is opened/closed in conjunction with the attachment and the removal of the toner cartridge T with respect to the process cartridge P. With the toner cartridge T attached to the process cartridge P, the toner is allowed to be discharged from the toner discharge port **61**. As described above, the development roller **32** of the process cartridge P develops the electrostatic latent image formed on the photosensitive drum **12** by the toner discharged from the toner discharge port **61**.

[0120] As shown in FIGS. **7** and **14**, the toner transport member **62** and the toner transport screw **63** are accommodated in the inside of the toner accommodation chamber **53**. The toner transport member **62** and the toner transport screw **63** can be referred to as a first rotation member of the toner cartridge T.

[0121] As shown in FIGS. **14** and **15**, the toner cartridge T has a toner discharge device **64** for discharging a toner from the toner discharge port **61**. The toner discharge device **64** has a pump portion **65** for feeding air by compression, an air guide **66** for guiding the air fed by the pump

portion **65** toward the toner discharge port **61**, a compression portion **67** for compressing the pump portion **65**, and an operation gear **68** for moving the compression portion **67**. The operation gear **68** includes an operation groove **68a** for moving the compression portion **67**.

[0122] The toner transport member **62** has a fin **62b** having flexibility, and a shaft **62c** to which the fin **62b** is attached, and rotates around the rotation axis **62a**, thereby transporting a toner toward the toner transport screw **63**. As shown in FIG. **15**, the fin **62b** comes in contact with the inner wall of the toner accommodation chamber **53**, to be deformed like a broken line from the natural state indicated with a solid line, for transporting the toner. The toner transport screw **63** has a spiral fin **63b1** and a screw shaft **63b2**, and rotates around the rotation axis **63a**, thereby transporting the toner toward the toner discharge port **61**. The spiral fin **63b1** and the screw shaft **63b2** are integrally formed.

[0123] On the other hand, the rotation of the operation gear **68** moves the compression portion **67** in engagement with the operation groove **68a** in the direction of the rotation axis of the operation gear **68**. The movement of the compression portion **67** compresses the pump portion **65**, so that the pump portion **65** feeds air toward the toner accommodation chamber **53**.

[0124] The fed air is guided toward the toner discharge port **61** by an air guide **66**. Incidentally, a cover member is arranged in the vicinity of the toner discharge port **61**, and the cover member covers a part of the toner transport screw **63**. As a result of this, the toner transported by the toner transport member **62** is suppressed from being discharged directly from the toner discharge port **61**, and the air released from the air guide **66** goes toward the toner discharge port **61** with efficiency.

[0125] Further, as shown in FIGS. **11A** and **11B**, the toner frame body **55** includes a toner receiving port (second opening) **84** for receiving the toner returned from the process cartridge P. The toner receiving port **84** establishes a communication between the outside of the toner frame body **55** and the toner collecting chamber **54**. The toner collecting chamber **54** accommodates the toner (collected toner) discharged from a toner return opening **20** shown in FIG. **8**, and having passed through the toner receiving port **84**.

[0126] In the axis direction LD, the toner receiving port **84** is arranged closer to the non-driving side of the toner frame body **55**. Namely, in the axis direction LD, the distance between the toner receiving port **84** and the toner non-driving end **55a2** is shorter than the distance between the toner receiving port **84** and the toner driving end **55a1**. Further, in the axis direction LD, the distance between the toner receiving port **84** and the toner non-driving end **55a2** is shorter than the distance between the toner receiving port **84** and the center **55a3** of the toner frame body **55**.

[0127] The toner receiving port **84** is covered with the receiving port shutter **85**. The receiving port shutter **85** is opened/closed in conjunction with the attachment and the removal of the toner cartridge T with respect to the process cartridge P.

[0128] More specifically, the receiving port shutter **85** is movable between the closing position at which the toner receiving port **84** is covered and the opening position at which the toner receiving port **84** is exposed. As shown in FIG. **8**, the attachment of the toner cartridge T to the process cartridge P brings the receiving port shutter **85** into contact with the process cartridge P, and causes the receiving port shutter **85** to move to the opening position. With the toner cartridge T attached to the process cartridge P, the toner receiving port **84** and the toner return opening **20** are opposed to each other.

[0129] As shown in FIG. **14**, the toner collecting chamber **54** includes a first collecting chamber **69**, a second collecting chamber **70**, and a third collecting chamber **71**. The toner cartridge T has a first collecting screw **72** for transporting a toner from the first collecting chamber **69** to the second collecting chamber **70**, and a second collecting screw (second rotation member) **73** for transporting a toner from the second collecting chamber **70** to the third collecting chamber **71**. The first collecting screw **72** and the second collecting screw **73** are accommodated in the toner collecting chamber **54**.

[0130] As shown in FIG. **8**, the first collecting screw **72** is rotatable around the rotation axis **72a**,

and the second collecting screw **73** is rotatable around the rotation axis **73a**. The direction of the rotation axis **72a** is in parallel with the axis direction LD, and the direction of the rotation axis **73a** is the direction crossing the axis direction LD. The angle formed between the direction orthogonal to the axis direction LD and the direction of the rotation axis **73a** is preferably smaller than the angle formed between the axis direction LD and the direction of the rotation axis **73a**, and the direction of the rotation axis **73a** is more preferably the direction orthogonal to the axis direction LD.

[0131] The toner received from the toner receiving port **84** is accumulated in the first collecting chamber **69**, and is transported from the first collecting chamber **69** to the second collecting chamber **70** by the first collecting screw **72**. The toner transported to the second collecting chamber **70** is transported upward in the vertical direction by the second collecting screw **73**, and is transported from the second collecting chamber **70** to the third collecting chamber **71**.

[0132] As shown in FIGS. **16A** and **16B**, the toner cartridge T has a second toner coupling **74**, a first toner coupling **75**, a transport gear **76**, and a discharge gear **77**.

[0133] The operation gear **68**, the second toner coupling **74**, the first toner coupling **75**, the transport gear **76**, and the discharge gear **77** are arranged on the driving side of the toner cartridge T. In other words, in the axis direction LD, the distance between the second toner coupling **74** or the first toner coupling **75**, and the toner driving end **55a1** is shorter than the distance between the second toner coupling **74** or the first toner coupling **75**, and the toner non-driving end **55a2**.

[0134] The driving end cover **57** shown in FIGS. **13** and **16A** covers at least a part of the operation gear **68**, the second toner coupling **74**, the first toner coupling **75**, the transport gear **76**, and the discharge gear **77**, and has a driving side toner guide **51**. The driving end cover **57** covers at least a part of the pump portion **65**.

[0135] The transport gear **76** is coupled with the toner transport member **62**, and the discharge gear **77** is coupled with the toner transport screw **63**. The second toner coupling **74** receives a driving force (external force) from the apparatus main body **100A**, and rotates the transport gear **76**. As a result, the toner transport member **62** is rotated. The first toner coupling **75** is in engagement with the operation gear **68** via an idler gear, and the operation gear **68** is in engagement with the discharge gear **77**. The first toner coupling **75** receives a driving force (external force) from the apparatus main body **100A**, and rotates the operation gear **68** and the discharge gear **77**. As a result, the pump portion **65** is compressed, so that the toner transport screw **63** is rotated.

[0136] In other words, the second toner coupling **74** has a function as a drive member for driving the toner transport member **62**. The first toner coupling **75** has a function as a drive member for driving the toner discharge device **64** and the toner transport screw **63**.

[0137] As shown in FIG. **17**, the toner cartridge T has a detection member **78** for detecting the toner amount in the toner collecting chamber **54**. The non-driving end cover **58** covers the detection member **78**, and has the non-driving side toner guide **52**.

[0138] The detection member **78** in the present Example is a light guide pair for guiding a light. The detection member **78** includes a light guide member **78a1** and a light guide member **78a2**, and is arranged on the non-driving side of the toner cartridge T. In other words, in the axis direction LD, the distance between the light guide member **78a1** or the light guide member **78a2**, and the toner non-driving end **55a2** is shorter than the distance between the light guide member **78a1** or the light guide member **78a2**, and the toner driving end **55a1**.

[0139] A part of the light guide member **78a1** and a part of the light guide member **78a2** are exposed at the toner collecting chamber **54**. One of the light guide member **78a1** and the light guide member **78a2** guides a light toward the inside of the toner collecting chamber **54** from the outside of the toner collecting chamber **54**. The light guided to the inside of the toner collecting chamber **54** passes through the toner collecting chamber **54**. The other of the light guide member **78a1** and the light guide member **78a2** guides a light to the outside of the toner collecting chamber **54** from the inside of the toner collecting chamber **54**.

[0140] The control portion **107** of the apparatus main body **100A** can detect the toner amount on the basis of the light that has passed through the inside of the toner collecting chamber **54** via the light guide member **78a1** and the light guide member **78a2**. Incidentally, as the detection member **78**, a mutually opposing electrode pair can be used. In this case, the control portion **107** of the apparatus main body **100A** can detect the toner amount on the basis of the change in electrostatic capacity between the electrode pair.

Process Cartridge

[0141] Referring to FIGS. **3** to **10**, the configuration of the process cartridge P in accordance with the present Example will be described in further details.

[0142] FIGS. **7** and **8** are each a cross sectional view of the process cartridge P and the toner cartridge T. Specifically, FIGS. **7** and **8** are each a cross sectional view of the process cartridge P and the toner cartridge T with the direction orthogonal to the axis direction LD assumed as the cross sectional direction. FIG. **8** is a view obtained by cutting the process cartridge P and the toner cartridge T along the rotation axis RS of a return screw **18** described later.

[0143] FIG. **9** is a side view for illustrating the driving transmission path of the process cartridge P. FIG. **10** is a view of the process cartridge P and the toner cartridge T as seen in the removal direction PDD.

[0144] The process cartridge P includes the development unit **30** and the drum unit **10**. The development unit **30** is movably (rotatably) coupled with respect to the drum unit **10**. As shown in FIGS. **5** and **6**, the process cartridge P has a driving side spring **37** and a non-driving side spring **38**, and the driving side spring **37** and the non-driving side spring **38** are attached to the drum unit **10** and the development unit **30**. The driving side spring **37** and the non-driving side spring **38** bias the development unit **30** so that the development roller **32** may be pressed against the photosensitive drum **12**.

[0145] As shown in FIGS. **7** and **8**, the development unit **30** has a development frame body **31**, a development roller **32** (a developer bearing member) for bearing a toner, a supply roller **33** (a supply member) for coming in contact with the development roller **32**, and supplying a toner, a development blade **34**, and a stirring unit **35** (a stirring member). The development frame body **31** supports the development roller **32**, the supply roller **33**, the development blade **34**, and the stirring unit **35**. The development frame body **31** is provided with a developer accommodation chamber **31a** and a development chamber **31b**.

[0146] The developer accommodation chamber **31a** is set at the top in the gravity direction of the development chamber **31b** via a wall surface portion **201e** that is a partition sheet shown in FIG. **1A**. Further, the wall surface portion **201e** is provided with an opening **201** for establishing a communication between the developer accommodation chamber **31a** and the development chamber **31b**.

[0147] In the developer accommodation chamber **31a**, a stirring unit **35** is arranged. In the development chamber **31b**, the development roller **32**, the supply roller **33**, and the development blade **34** are arranged.

[0148] The stirring unit **35** includes a rotatively supported stirring shaft **35c**, and a flexible sheet member **35a** fixed to the stirring shaft **35c**.

[0149] As shown in FIG. **7**, the supply roller **33** rotates in the same direction as that of the development roller **32** (an arrow K direction), and the stirring unit **35** rotates in the opposite direction to that of the supply roller **33** (an arrow Q direction). Further, the longitudinal direction of the supply roller **33** in the development unit **30** will be hereinafter referred to as a longitudinal direction.

[0150] The toner supplied from the toner cartridge T is accommodated in the developer accommodation chamber **31a**. The stirring unit **35** transports the toner accommodated in the developer accommodation chamber **31a** to the development chamber **31b**. The toner transported to the development chamber **31b** is supplied to the development roller **32** by the supply roller **33** that

comes in contact with the development roller 32 to be rotated. The toner supplied to the development roller 32 is regulated by the development blade 34, so that a toner layer is formed on the surface of the development roller 32. The development blade 34 has a function as a layer thickness regulating member for regulating the thickness of the toner layer.

[0151] As shown in FIGS. 7 and 8, the drum unit 10 has a drum frame body 11, a photosensitive drum 12 (image bearing member), a charging roller 13, a cleaning blade 14, an intermediate transport member 15, an intermediate screw 16, a transmission shaft 17, and a return screw 18. The drum frame body 11 supports the photosensitive drum 12, the charging roller 13, the cleaning blade 14, the intermediate transport member 15, the intermediate screw 16, the transmission shaft 17, and the return screw 18. Further, the drum unit 10 has a memory tag 90P described later.

[0152] The drum frame body 11 includes a cleaning collecting chamber 19. In the cleaning collecting chamber 19, the intermediate transport member 15, the intermediate screw 16, and the return screw 18 are arranged.

[0153] As shown in FIG. 8, the drum frame body 11 includes a return path 45 for accommodating the return screw 18. The return path 45 can be said to be a part of the cleaning collecting chamber 19.

[0154] The charging roller 13 is in contact with the photosensitive drum 12, and is rotated by the photosensitive drum 12. The cleaning blade 14 is in contact with the photosensitive drum 12, and collects the toner left on the surface of the photosensitive drum 12. The collected toner (the waste toner, the residual toner, or the recovered toner) is accommodated in the cleaning collecting chamber 19. The collected toner is transported toward the intermediate screw 16 by the intermediate transport member 15, and the intermediate screw 16 transports the collected toner toward the return screw 18. The intermediate transport member 15 transports the collected toner toward the direction crossing the axis direction LD. The intermediate screw 16 transports the collected toner along the axis direction LD.

[0155] The return screw (rotation member) 18 rotates around the rotation axis (the second axis) RS shown in FIG. 6. The direction of extension of the rotation axis RS of the return screw 18 is referred to as the rotation axis direction (the second direction) of the return screw 18.

[0156] The rotation axis direction of the return screw 18 is the direction crossing the axis direction LD. The angle formed between the direction orthogonal to the axis direction LD and the rotation axis direction of the return screw 18 is preferably smaller than the angle formed between the axis direction LD and the rotation axis direction of the return screw 18, and the rotation axis direction of the return screw 18 is more preferably the direction orthogonal to the axis direction LD.

[0157] As shown in FIG. 8, the drum frame body 11 includes a return opening 20. The return opening 20 establishes a communication between the return path 45 of the cleaning collecting chamber 19 and the outside of the drum frame body 11, and is opposed to the return screw 18. The collected toner delivered to the return screw 18 from the intermediate screw 16 is transported toward the return opening 20 by the return screw 18 and is discharged from the return opening 20, to pass through a toner receiving port 84 described later, and to be received by the toner cartridge T.

[0158] Thus, the return screw 18 has a function as a transport member for transporting the toner collected from the photosensitive drum 12 toward the toner receiving port 84. The direction of transport of the collected toner by the return screw 18 is the direction toward the toner cartridge T from the process cartridge P, and the upward direction in the vertical direction.

[0159] The return screw 18 has the spiral fin 18b and the screw shaft 18c, and rotates around the rotation axis RS, thereby transporting the toner toward the return opening 20. The spiral fin 18b and the screw shaft 18c are integrally formed.

[0160] As shown in FIGS. 3 and 5, the process cartridge P has a process coupling 36 (a first input portion or a development drive member), and a drum gear 21 (a second input portion or a drum drive member). The engagement of the process coupling 36 with the main body coupling of the apparatus main body 100A results in the transmission of a driving force (an external force) from the

apparatus main body **100A** to the process coupling **36**. The engagement of the drum gear **21** with the main body gear of the apparatus main body **100A** results in the transmission of a driving force (an external force) from the apparatus main body **100A** to the drum gear **21**, so that the drum gear **21** rotates. The rotation of the drum gear **21** drives and rotates the photosensitive drum **12**.

[0161] In the present Example, the process coupling **36** and the drum gear **21** are arranged on the driving side of the process cartridge P. Namely, in the axis direction LD, the distance between the process driving end **11f1** and the process coupling **36** is shorter than the distance between the process non-driving end **11f2** and the process coupling **36**. Similarly, in the axis direction LD, the distance between the process driving end **11f1** and the drum gear **21** is shorter than the distance between the process non-driving end **11f2** and the drum gear **21**.

[0162] As shown in FIG. **9**, the process cartridge P has a stirring gear **39** for driving the stirring unit **35**, a development gear **40** for driving the development roller **32**, and a supply gear **41** for driving the supply roller **33**. The stirring gear **39**, the development gear **40**, and the supply gear **41** are coupled to the process coupling **36** via a plurality of the idler gears **42**. The rotation of the process coupling **36** drives and rotates the development roller **32**, the supply roller **33**, and the stirring unit **35**.

[0163] Further, the process cartridge P has an intermediate transport gear **24** for driving the intermediate transport member **15**, an intermediate screw gear **25** for driving the intermediate screw **16**, and a shaft gear **26** for driving the transmission shaft **17**. The intermediate transport gear **24**, the intermediate screw gear **25**, and the shaft gear **26** are coupled to the process coupling **36** via a plurality of the idler gears **27**. The rotation of the process coupling **36** rotates the intermediate transport member **15**, the intermediate screw **16**, and the transmission shaft **17**.

[0164] As shown in FIG. **8**, the return path **45** and the return screw **18** are arranged on the non-driving side of the process cartridge P. In other words, in the axis direction LD, the distance between the return screw **18** and the process non-driving end **11f2** is shorter than the distance between the return screw **18** and the process driving end **11f1**.

[0165] More specifically, as shown in FIG. **10**, in the axis direction LD, the distance between the rotation axis RS of the return screw **18** and the process non-driving end **11f2** is shorter than the distance between the rotation axis RS and the process driving end **11f1**. Whereas, in the axis direction LD, the distance between the rotation axis RS and the process non-driving end **11f2** is shorter than the distance between the rotation axis RS and the center **11f3** of the drum frame body **11**.

[0166] As shown in FIG. **8**, the transmission shaft **17** is attached to the transmission gear **28**, and a return gear **29** to be in engagement with the transmission gear **28** is attached to the return screw **18**. The transmission gear **28** and the return gear **29** are bevel gears, so that the rotation of the transmission shaft **17** rotates the return screw **18**. Namely, the driving force transmitted to the process coupling **36** is transmitted from the driving side of the process cartridge P to the non-driving side of the process cartridge P by the transmission shaft **17**, and is transmitted to the return screw **18**. In other words, the process coupling **36** is configured so as to drive the return screw **18**.

[0167] As shown in FIG. **10**, the transmission shaft **17** is arranged outside the cleaning collecting chamber **19**, and the transmission gear **28** and the return gear **29** are in engagement with each other outside the cleaning collecting chamber **19**.

[0168] As shown in FIGS. **4** and **6**, a development roller electrode **32a** (a development roller contact), a development blade electrode **34a** (a development blade contact), a supply roller electrode **33a** (a supply roller contact) and a charging roller electrode **13a** (a charging contact) are arranged on the non-driving side of the process cartridge P. In other words, the distance between the process non-driving end **11f2** and the charging roller electrode **13a** is shorter than the distance between the process driving end **11f1** and the charging roller electrode **13a**. Similarly, the distance between the process non-driving end **11f2**, and the development roller electrode **32a**, the development blade electrode **34a**, or the supply roller electrode **33a** is shorter than the distance

between the process driving end **11/1**, and the development roller electrode **32a**, the development blade electrode **34a**, or the supply roller electrode **33a**.

[0169] The development roller electrode **32a**, the development blade electrode **34a**, the supply roller electrode **33a**, and the charging roller electrode **13a** are electrically connected with the development roller **32**, the development blade **34**, the supply roller **33**, and the charging roller **13**, respectively. When an image formation operation is performed, the development roller electrode **32a**, the development blade electrode **34a**, the supply roller electrode **33a**, and the charging roller electrode **13a** are applied with a prescribed voltage from the power supply portion **170** of the apparatus main body **100A**.

[0170] Each material for the development roller electrode **32a**, the development blade electrode **34a**, the supply roller electrode **33a**, and the charging roller electrode **13a** may be a metal, or may be a resin having conductivity.

[0171] Incidentally, in the present Example, the direction of each rotation axis of the development roller **32**, the supply roller **33**, the stirring unit **35**, the charging roller **13**, the intermediate transport member **15**, the intermediate screw **16**, and the transmission shaft **17** is in parallel with the axis direction LD. The direction of the rotation axis of each gear except for the process coupling **36**, the drum gear **21**, and the return gear **29** is also in parallel with the axis direction LD.

With Regard to Detailed Configuration of Inside of Development Unit **30** Characteristic of the Present Invention

[0172] Referring to FIGS. **1A** and **1B**, a description will be given to the shape of the opening **201** of the development unit **30** in accordance with the present invention. FIG. **1A** is a side surface cross sectional view of the development unit showing the shape of the opening **201** at the end in the longitudinal direction. FIG. **1B** is a side surface cross sectional view of the development unit showing the shape of the opening **201** at the central portion in the longitudinal direction.

[0173] As shown in FIG. **1A**, in the cross sectional view of the end in the longitudinal direction, the opening **201** is formed by being partitioned by a second end **201b** situated on the upstream side in a rotation direction Q of the stirring unit **35**, and a fourth end **201d** situated on the downstream side in the rotation direction Q of the stirring unit **35**. The second end **201b** is the opening end on the upstream side at the end in the longitudinal direction, and the fourth end **201d** is the opening end at the end on the downstream side in the longitudinal direction.

[0174] As shown in FIG. **1B**, in the cross sectional view of the central portion in the longitudinal direction, the opening **201** is formed by being partitioned by a first end **201a** situated on the upstream side in the rotation direction Q of the stirring unit **35**, and a third end **201c** situated on the downstream side in the rotation direction Q of the stirring unit **35**. The first end **201a** is the opening end on the upstream side at the central portion in the longitudinal direction, and the third end **201c** is the opening end on the downstream side at the central portion in the longitudinal direction.

[0175] The length of the sheet member **35a** of the stirring unit **35** will be described. The distance between the rotation axis center of the stirring unit **35** and the free tip portion of the sheet when the sheet member **35a** has no flexure is referred to as the “free distance of sheet”. Whereas, the distance between the rotation center of the stirring unit **35** and the first end **201a** is referred to as the “center-end distance at central portion”. Further, the distance between the rotation center of the stirring unit **35** and the second end **201b** is referred to as the “center-end distance at end”. At this step, the “free distance of sheet” is longer than the “center-end distance at central portion” and the “center-end distance at end”. Therefore, the sheet member **35a** of the stirring unit **35** is configured to come in contact with the wall surface portion **201e** in the vicinity of the opening **201** of the developer accommodation chamber **31a** while bending, and to feed a developer of the developer accommodation chamber **31a** into the development chamber **31b** with stirring.

[0176] Herein, the contact region of the supply roller **33** with the development roller **32** is referred to as a contact region **202**. As shown in the cross sectional view, when the supply roller **33** bites into the development roller **32** by a prescribed distance, the contact region **202** may be at the

position of the center of the biting region, may be the end, or may be at other prescribed positions. At this step, the position of the first end **201a** of the opening **201** is set on the side closer to the contact region **202** of the supply roller **33** than the second end **201b** of the opening **201**. Specifically, the relationship of “distance L1 between first end **201a** and contact region **202**” < “distance L2 between second end **201b** and contact region **202**” holds. At this step, the distance L1 and the distance L2 are not limited to the distance between the end and the contact region **202**, but may be the distance between the end and the inlet portion **202a**, or the distance between the end and an outlet portion **202b**.

[0177] Further, in FIG. 1A, for measuring the distance L2, the distance between the line passing through the second end **201b**, and perpendicular to the wall surface portion **201e**, and the line passing through the contact region **202**, and perpendicular to the wall surface portion **201e** is measured. Whereas, in FIG. 1B, for measuring the distance L1, the distance between the line passing through the first end **201a**, and perpendicular to the wall surface portion **201e**, and the line passing through the contact region **202**, and perpendicular to the wall surface portion **201e** is measured. However for example, not the line perpendicular to the wall surface portion **201e**, but the line connecting the second end **201b** (or the first end **201a**) and the rotation axis center of the stirring unit **35** may be used. Alternatively, the distance between the second end **201b** (or the first end **201a**) and the contact region **202** may be directly measured.

Additional Configuration

[0178] In addition to the configurations up to this point, further, in the present Example, each of the first end **201a** and the second end **201b** is set at a more effective position. Below, a description will be given by reference to the side surface cross sectional view of each development unit shown in FIGS. 18A and 18B. FIG. 18A shows the shape of the opening **201** at each opposite end in the longitudinal direction, and FIG. 18B shows the shape of the opening **201** at the central portion in the longitudinal direction.

[0179] As shown in FIGS. 18A and 18B, the line connecting the stirring shaft center and the supply roller center is referred to as L3. As shown in FIG. 18B, the first end **201a** of the opening **201** is situated downstream of L3 in the rotation direction Q of the stirring unit **35**. On the other hand, as shown in FIG. 18A, the second end **201b** of the opening **201** is situated upstream of L3 in the rotation direction Q of the stirring unit **35**.

[0180] Further, the size of the opening **201** in the present Example will be described by reference to FIG. 20. FIG. 20 is a view of the opening **201** as seen in the front direction from the inside of the developer accommodation chamber **31a**. In the present Example, as shown in FIG. 20, the opening **201** satisfies the relationship of “distance L4 between first end **201a** and third end **201c**” < “distance L5 between second end **201b** and fourth end **201d**”.

[0181] Herein, the shape of the opening **201** may be the shape as shown in FIGS. 22A and 22B. FIGS. 22A and 22B are each a view of the opening **201** as seen in the front direction from the inside of the developer accommodation chamber **31a**, showing another shape example of the opening **201**.

[0182] The shape without a partition of the opening **201** in the longitudinal direction as shown in FIG. 22A, or the shape in which the first end **201a** and the second end **201b** of the opening **201** continuously change in the longitudinal direction as shown in FIG. 22B is also acceptable. Further, the shape of the opening **201** is not limited to the examples so long as it satisfies the positional relationship previously described.

Description of Circulation of Developer Inside Development Unit 30

[0183] Below, referring to FIGS. 19A and 19B, a description will be given to the circulation of the developer inside the development unit **30**. FIG. 19A is a cross sectional view of the development unit **30** as seen in the longitudinal direction. FIG. 19B is a cross sectional view at the line along the longitudinal direction, and shows the movement of the developer in the longitudinal direction in the development chamber **31b** in the periphery of the supply roller **33** of the development unit **30**.

[0184] As shown in FIG. 19A, the developer to be accommodated in the developer accommodation chamber 31a has its own weight applied in an arrow Z direction, and hence it becomes more compact as it moves toward the development chamber 31b. Further, the sheet member 35a of the stirring unit 35 comes in contact with the wall surface portion 201e of the opening 201, and rotates in an arrow Q direction while bending. As a result, the developer in the developer accommodation chamber 31a is fed in an arrow R direction with stirring inside the accommodation chamber. Particularly, at the central portion in the longitudinal direction, the amount of the developer to be fed from the developer accommodation chamber 31a to the development chamber 31b tends to be larger than at the end. Incidentally, in FIG. 19A, the tip (the end on the side not coupled with the shaft) of the sheet member 35a moves while being in contact with the opening 201 when passing through the vicinity of the opening 201. However, the length of the sheet member 35a may be shorter than this so long as the developer can be stirred. The length of the sheet member 35a may be just large enough that the tip does not come in contact with the opening 201 when it passes through the vicinity of the opening.

[0185] On the other hand, the developer in the development chamber 31b is included in the supply roller 33 from the inlet portion 202a. The developer included in the supply roller 33 is discharged from the outlet portion 202b, thereby circulating the developer in the development chamber 31b. The action of the supply roller 33 is referred to as the “breath of the supply roller 33”. Incidentally, in the present Example, in order to show the action of the supply roller 33 and the movement of the developer, a description has been given by distinguishing the contact region 202, the inlet portion 202a, and the outlet portion 202b. However, in actuality, respective locations are not clearly separable. For this reason, for describing the positional relationship and the acting direction in the present invention, these are not supposed to be distinguished and limited.

[0186] Further, referring to FIG. 19B, a description will be given to the movement of the developer in the longitudinal direction of the supply roller in the development chamber 31b. Of the developers in the development chamber 31b, the developer that was not included in the supply roller 33 in the vicinity of the inlet portion 202a by the rotation of the supply roller 33 is fed to the longitudinal end of the process cartridge (in the arrow S direction).

[0187] As described up to this point, the developer is circulated, and generally, the circulation of the developer is more preferably performed smoothly. However, an increase in weight of the developer in the developer accommodation chamber 31a results in the compaction of the lower part in the gravity direction in the developer accommodation chamber 31a. As a result, the amount of the developer to be fed to the development chamber 31b increases. Particularly, at the central portion in the longitudinal direction, the effect tends to increase. Then, in the development chamber 31b, the previous developer fed from the developer accommodation chamber 31a compacts the vicinity of the outlet portion 202b with the developer. This makes the “breath of the supply roller” impossible. As a result, the amount of the developer accumulated in the vicinity of the inlet portion 202a to be fed to the longitudinal end of the process cartridge (in the arrow S direction) by the rotation of the supply roller 33 increases.

[0188] Then, at the end in the longitudinal direction of the development chamber 31b, the circulation of the developer is less. For this reason, the developer becomes compact. As a result, an image defective due to toner deterioration may be caused.

Effect of the Present Invention

[0189] In consideration of the circulation of the developer described above, as shown in FIGS. 1A and 1B, the satisfaction of the relationship of distance $L1 < \text{distance } L2$ can reduce the amount of the developer to be transported into the development chamber 31b from the developer accommodation chamber 31a at the central portion in the longitudinal direction. As a result, it is possible to make proper the amount of the developer staying in the vicinity of the inlet portion 202a of the supply roller 33 in the development chamber 31b. For this reason, it is possible to suppress the amount of the developer to be fed to each longitudinal end of the supply roller 33 in the development chamber

31b (the arrow S direction (see FIGS. **19A** and **19B**)) by the rotation of the supply roller **33**.

[0190] Further, as shown in FIG. **18B**, at the central portion in the longitudinal direction, the position of the first end **201a** is set at downstream in the rotation direction of the line L3. With this configuration, at the central portion in the longitudinal direction, the direction in which the transport force F1 of the developer by the sheet member **35a** of the stirring unit **35** acts can be shifted from the direction toward the outlet portion **202b**. For this reason, it is possible to more effectively reduce the amount of the developer to be transported into the development chamber **31b** from the developer accommodation chamber **31a** by the transport force of the stirring unit **35** at the longitudinal central portion of the supply roller.

[0191] On the other hand, as shown in FIG. **18A**, at the opposite ends in the longitudinal direction, the position of the second end **201b** is set at upstream of the line L3 in the rotation direction. With this configuration, the direction in which the transport force F2 of the tip portion **35b** of the bent sheet member **35a** acts can be turned in the direction of the outlet portion **202b** of the supply roller **33**. Further, the tip portion **35b** of the sheet member **35a** can enter the inside of the development chamber **31b**. As a result, the developer can be effectively circulated in the development chamber **31b** at both ends in the longitudinal direction.

[0192] Further, as shown in FIG. **20**, by satisfying the relationship of distance L4 < distance L5 for the size of the opening **201**, it is possible to reduce the amount of the developer to be transported into the development chamber **31b** from the developer accommodation chamber **31a** at the central portion in the longitudinal direction to which the amount of the developer to be fed tends to be large. For this reason, it is possible to make the circulation of the whole development unit proper.

[0193] Up to this point, by achieving the configuration of the present Example, it is possible to provide a development unit that can make proper the circulation balance of the developer in the development unit, and that does not undergo the leakage of the developer or an image detriment due to a circulation failure.

[0194] Incidentally, the present invention is not limited to the replenishment configuration including the process cartridge and the toner cartridge previously described. As shown in FIGS. **21A** and **21B**, the same effect is also expected for an integral type cartridge including the process cartridge and the toner cartridge integrally. FIGS. **21A** and **21B** are each a cross sectional view showing the shape of the opening **201** with an all-in-one cartridge configuration in which the developer accommodation chamber **31a** is filled with a large volume of a developer, including the same configuration as that of the present Example therein. Such an integral type configuration can also provide the effect of making the circulation of the developer proper.

[0195] As described up to this point, with the configuration described in the present specification, in the development unit configuration in which the developer accommodation chamber having a stirring unit is set at the top in the gravity direction of the development chamber, even when the weight of the developer increases, it is possible to keep the circulation of the developer proper. Namely, the opening coupling the developer accommodation chamber and the development chamber is in the shape that allows the developer to pass therethrough more easily at the ends in the longitudinal direction than at the central portion. Thus, by making the position and the shape of the opening different between at the ends in the longitudinal direction and at the central portion, it is possible to properly circulate the developer. Typically, for example, as described in FIGS. **20**, **22A**, and **22B**, it is configured such that the frontage is wider at each end in the longitudinal direction than at the central portion. However, it is essential only that the shape of the opening is the shape that allows the developer to pass more easily at each end in the longitudinal direction than at the central portion. For example, it is only required that a part of the end has a portion with a larger frontage than that of the central portion.

[0196] While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all

such modifications and equivalent structures and functions.

[0197] This application claims the benefit of Japanese Patent Application No. 2024-033700, filed on Mar. 6, 2024, which is hereby incorporated by reference wherein in its entirety.

Claims

1. A developing apparatus comprising: a developer bearing member for bearing a developer; a supply member for coming in contact with the developer bearing member, thereby forming a contact region, and supplying the developer to a surface of the developer bearing member in the contact region; a development chamber including the developer bearing member and the supply member arranged therein; a developer accommodation chamber for accommodating the developer; and a stirring member having a shaft, and a sheet member held by the shaft, the sheet member stirring the developer accommodated in the developer accommodation chamber by rotation of the shaft, wherein, the developer accommodation chamber is set at a top in a gravity direction of the development chamber in a posture in which the developing apparatus is mounted in an image forming apparatus, the developer bearing member, the supply member, and the stirring member are rotatable around a rotation axis along a longitudinal direction of the developing apparatus as a center, a wall surface portion for partitioning the developer accommodation chamber and the development chamber is provided with an opening for enabling a movement of the developer from the developer accommodation chamber to the development chamber along the longitudinal direction, and in a cross section in the longitudinal direction, the opening has an end on an upstream side and an end on a downstream side in a direction of passage of the sheet member according to rotation of the stirring member, and a distance between a first end and a contact region of the developer bearing member and the supply member is shorter than a distance between a second end and the contact region where the first end represents the end on the upstream side at a central portion in the longitudinal direction, and the second end represents the end on the upstream side at the end in the longitudinal direction.
2. The developing apparatus according to claim 1, wherein in a cross section in the longitudinal direction, the first end is situated on a downstream side in a rotation direction of the stirring member than a line connecting a rotation center of the shaft and a rotation center of the supply member, and the second end is situated on an upstream side in the rotation direction of the stirring member than the line connecting a rotation center of the shaft and the rotation center of the supply member.
3. The developing apparatus according to claim 1, wherein the sheet member of the stirring member has flexibility, a distance between a rotation center of the shaft and a free tip portion of the sheet member when the sheet member has no flexure is longer than a distance between the rotation center and the wall surface portion, and the sheet member rotates while coming in contact with the wall surface portion.
4. The developing apparatus according to claim 3, wherein the sheet member can enter the opening at an end in the longitudinal direction, and does not enter the opening at a central portion in the longitudinal direction.
5. The developing apparatus according to claim 1, wherein in a cross section in the longitudinal direction, a distance between the first end at a central portion in the longitudinal direction and a third end is shorter than a distance between the second end at an end in the longitudinal direction and a fourth end where the third end represents an end on the downstream side at a central portion in the longitudinal direction of the opening, and the fourth end represents an end on the downstream side at an end in the longitudinal direction.
6. An image forming apparatus comprising: a developer bearing member for bearing a developer; a supply member for coming in contact with the developer bearing member, thereby forming a contact region, and supplying the developer to a surface of the developer bearing member in the

contact region; a development chamber including the developer bearing member and the supply member arranged therein; a developer accommodation chamber for accommodating the developer; a stirring member having a shaft, and a sheet member held by the shaft, the sheet member stirring the developer accommodated in the developer accommodation chamber by rotation of the shaft; and an image bearing member for developing an electrostatic latent image formed on a surface when the developer is supplied from the developer bearing member, wherein, the developer accommodation chamber is set at a top in a gravity direction of the development chamber, the developer bearing member, the supply member, and the stirring member are each rotatable around a rotation axis along a longitudinal direction as a center, a wall surface portion for partitioning the developer accommodation chamber and the development chamber is provided with an opening for enabling a movement of the developer from the developer accommodation chamber to the development chamber along the longitudinal direction, and in a cross section in the longitudinal direction, the opening has an end on an upstream side and an end on a downstream side in a direction of passage of the sheet member according to rotation of the stirring member, and a distance between a first end and a contact region of the developer bearing member and the supply member is shorter than a distance between a second end and the contact region where the first end represents the end on the upstream side at a central portion in the longitudinal direction, and the second end represents the end on the upstream side at the end in the longitudinal direction.

7. A developing apparatus comprising: a developer bearing member for bearing a developer; a supply member for coming in contact with the developer bearing member, thereby forming a contact region, and supplying the developer to a surface of the developer bearing member in the contact region; a development chamber including the developer bearing member and the supply member arranged therein; a developer accommodation chamber for accommodating the developer; and a stirring member having a shaft, and a sheet member held by the shaft, the sheet member stirring the developer accommodated in the developer accommodation chamber by rotation of the shaft, wherein, the developer accommodation chamber is set at a top in a gravity direction of the development chamber in a posture in which the developing apparatus is mounted in an image forming apparatus, the developer bearing member, the supply member, and the stirring member are each rotatable around a rotation axis along a longitudinal direction of the developing apparatus as a center, a wall surface portion for partitioning the developer accommodation chamber and the development chamber is provided with an opening for enabling a movement of the developer from the developer accommodation chamber to the development chamber along the longitudinal direction, and in a cross section in the longitudinal direction, the opening has an end on an upstream side and an end on a downstream side in a direction of passage of the sheet member according to rotation of the stirring member, and a width between the end on the upstream side and the end on the downstream side at a central portion in the longitudinal direction is narrower than a width between the end on the upstream side and the end on the downstream side at the ends in the longitudinal direction.
